Sorathorn Thongpitukthavorn

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Education

Master of Engineering in Computer Science, *Cornell University (Cornell Tech)* 08/2026 – 05/2027 | New York, NY
Coursework: Applied Machine Learning, Optimization Methods, Networks & Markets, Product Studio.

Bachelor of Arts in Computer Science, *Boston University* 09/2022 – 05/2026 | Boston, MA
GPA: 3.61/4.00

Work Experience

Software & Automation Engineer (Contractual), *Asia Inter Rice Co., Ltd.* 05/2024 – 08/2026 | Bangkok, Thailand

- Automated purchase-order ETL across 11 retailers using Electron, Flask, and Pandas, standardizing PDF, Excel, HTML, and XML feeds into ERP records to cut processing time by >90%.
- Shipped a containerized property platform (React, FastAPI, SQLite) on GCP with Docker Compose and blue-green deployments for 60+ units, reducing late payments by 80% via LINE webhooks.
- Engineered a Google Cloud Vision OCR pipeline to extract utility-meter readings directly from tenant photos, automating billing calculations and saving 20 hours/week of manual entry.
- Built a multi-bank reconciliation desktop app (Electron, FastAPI, pdfplumber) with 4 automated workflows and zero-lock shared SQLite concurrency to match remittances to open invoices.

Software Developer Intern, *Datapro Computer Systems* 05/2025 – 08/2025 | Bangkok, Thailand

- Developed a passport-processing mobile app (React Native, TypeScript, Spring Boot, PostgreSQL) with pdfplumber OCR validation, deploying via Docker onto on-premise client servers.
- Architected containerized CI/CD pipelines with Docker and GitLab, cutting deployment cycle times by 40% while enforcing strict staging-to-production environment parity.
- Automated end-to-end integration and regression testing using Selenium and Jira, resolving 30+ critical bugs and performance bottlenecks prior to User Acceptance Testing

Technical Projects

Skail — Multi-Agent Autonomous Coding Harness , *Python, LangGraph, DeepAgents, SQLite, Git*

- Architected an open-source, local-first coding harness orchestrating a lead agent and 3 concurrent subagents (Explorer, Implementer, Tester) via typed, dependency-aware DAG task graphs.
- Isolated concurrent file-writing agents in Git worktrees with serialized change-set merging, deterministic loop-stagnation detection, and automated test/build verification gates.
- Implemented capability-floor LLM routing with transactional budget reservations and a high-throughput SQLite journal supporting crash recovery and human-in-the-loop interrupts.

Project SCYLLA — Quantitative Options & Anomaly Terminal , *C++, Crow, Python, FastAPI, LightGBM*

- Built an asynchronous dual-engine analytics platform pairing a multithreaded C++ Crow microservice and native LightGBM C-API inference engine with Python feature-engineering pipelines.
- Engineered real-time market anomaly detection to flag institutional block flow and regime shifts, backed by a jittered retry pipeline with circuit-breaker fault tolerance.
- Trained 5 LightGBM quantile models on options flow and volatility, validating signal execution via chronological walk-forward backtesting and dynamic Kelly criterion position sizing.

Cloud Infrastructure & Distributed Data Pipelines , *Docker, GKE (Kubernetes), Apache Beam, Dataflow, GCP*

- Containerized microservices on Google Kubernetes Engine (GKE) behind multi-zone Layer 4 Load Balancers, achieving high availability and <3s failover recovery.
- Engineered distributed Apache Beam pipelines on Cloud Dataflow to process 10M+ records, integrating event-driven Pub/Sub and Cloud Functions for automated security alerting.
- Automated reproducible provisioning of Compute Engine VMs, Cloud SQL (MySQL), and least-privilege IAM service accounts using Bash and gcloud CLI scripts.

Technical Skills

- Languages: Python, C++, TypeScript, JavaScript, SQL, Java, Go, C, Bash
- AI & Machine Learning: LangGraph, DeepAgents, Multi-Agent Orchestration, LiteLLM, LightGBM, PyTorch, Scikit-learn, OpenAI API
- Backend & Cloud Infrastructure: FastAPI, Flask, Spring Boot, React, React Native, Electron, Docker, GKE (Kubernetes), Apache Beam, GCP, AWS, Linux, REST APIs, WebSockets
- Databases & Developer Tools: PostgreSQL, SQLite, Cloud SQL (MySQL), Git, GitLab CI/CD, GitHub Actions, Selenium, Pandas, openpyxl, pdfplumber
- Certifications: CompTIA Security+, CompTIA Network+, CompTIA A+